

Drosophila Courtship Behavior Analysis Pipeline

Technical Documentation

A Python-Based Computer Vision System for Automated
Detection and Classification of Fruit Fly Courtship Behaviors
in Multi-Chamber Arena Experiments

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1. Motivation and Background

1.1 The Problem

Drosophila melanogaster courtship behavior constitutes one of the most extensively studied innate behavioral repertoires in genetics and neuroscience. Male flies perform a stereotyped sequence of actions directed toward females: orientation toward the target, following, unilateral wing extension to produce a species-specific courtship song, circling, licking, attempted copulation, and copulation (Hall, 1994; Greenspan and Ferveur, 2000). Each of these behavioral elements is under genetic control and can be selectively disrupted by mutations in genes such as fruitless (*fru*) and doublesex (*dsx*) (Yamamoto and Koganezawa, 2013).

Quantification of courtship behavior is essential for phenotyping mutant lines, assessing the effects of neural circuit manipulations, and studying the sensory ecology of mate choice. The standard metric, the Courtship Index (CI), was introduced by Siegel and Hall (1979) and is defined as the fraction of observation time during which the male engages in any courtship-related activity. Historically, scoring courtship has been performed manually by trained observers viewing recorded videos, a process that is labor-intensive, subject to inter-observer variability, and poorly scalable to high-throughput experimental designs.

1.2 Existing Approaches and Their Limitations

Several automated systems have been developed to address this bottleneck. Ctrax (Branson et al., 2009) provides robust multi-fly tracking using background subtraction and ellipse fitting but does not classify behaviors. FlyTracker (Eyjolfsson et al., 2014) extends tracking to include social feature computation for pairs of interacting flies. JAABA (Kabra et al., 2013) builds upon these trackers by providing interactive machine-learning classifiers trained via GentleBoost (Friedman et al., 2000) on windowed per-frame features.

While powerful, these tools have practical limitations. They typically require MATLAB licenses, involve multi-step workflows across separate applications, and demand researcher-labeled training data for each new experimental setup. For laboratories running routine courtship assays with standardized arenas, a self-contained system that integrates tracking, feature extraction, and rule-based classification into a single pipeline would be of considerable practical value.

1.3 Design Goals

This system was designed with the following objectives:

- Provide an end-to-end pipeline that accepts a raw multi-chamber arena video and produces per-chamber courtship annotations and summary statistics with no manual intervention.
- Support both heuristic (rule-based) classification using literature-derived thresholds and JAABA-trained boosting classifiers for laboratories that have existing .jab model files.
- Run entirely in Python with open-source dependencies, requiring no MATLAB license.
- Expose a web-based interface (Streamlit) that allows parameter tuning, chamber selection, and visual verification of results.
- Produce output compatible with downstream statistical analysis, including per-bout CSVs and summary statistics with Courtship Index, latency metrics, and sex inference.

2. System Architecture

2.1 Pipeline Overview

The analysis pipeline processes a single multi-chamber arena video through five sequential stages:

Stage 1: Chamber Detection. The first frame (or a median of several frames for robustness to occlusions) is analyzed to locate individual chambers within the arena grid. Detected regions are snapped to a regular grid to ensure uniform, edge-inclusive bounding boxes.

Stage 2: Video Preprocessing. Each chamber is cropped from the source video. An optional stabilization step uses Enhanced Correlation Coefficient (ECC) alignment (Evangelidis and Psarakis, 2008) to correct for camera vibration or arena drift.

Stage 3: Fly Tracking. Within each cropped chamber video, two flies are tracked frame-by-frame using MOG2 background subtraction (Zivkovic, 2004; Zivkovic and van der Heijden, 2006), contour detection, and ellipse fitting (Fitzgibbon et al., 1999). Frame-to-frame identity is maintained using the Hungarian algorithm (Kuhn, 1955; Munkres, 1957).

Stage 4: Feature Extraction and Behavior Classification. Kinematic and social geometry features are computed from the tracking data. Behaviors are classified either through rule-based heuristics with literature-derived thresholds or through JAABA boosting classifiers.

Stage 5: Export. Per-chamber CSV files are generated containing bout-level annotations (behavior, fly identity, start/end times, duration) along with summary statistics (Courtship Index, latency to first courtship, latency to copulation).

2.2 Module Organization

The codebase is organized into the following modules:

Module	Description
app.py	Streamlit UI, orchestration, parallel processing
tracking/tracker.py	MOG2 segmentation, ellipse fitting, identity assignment
tracking/features.py	Kinematic/social features, JAABA window features
classification/heuristic.py	Rule-based behavior classifier (5 behaviors)
classification/inference.py	Boosting classifier evaluation, bout detection
classification/jab_parser.py	JAABA .jab file parser (v7 and v7.3 HDF5)
utils/video.py	Chamber detection, cropping, ECC stabilization
utils/export.py	CSV export, sex inference, summary statistics

3. Chamber Detection and Video Preprocessing

3.1 Multi-Panel Grid Detection

Multi-chamber courtship arenas typically consist of one or two grid panels, each containing a regular array of small rectangular chambers. The arena may be imaged with varying numbers of chambers and spatial layouts. Chamber detection proceeds hierarchically: first, large grid regions are identified, then individual chambers are detected within each region.

To mitigate transient occlusions (e.g., a researcher's hand entering the frame during video setup), the system samples multiple frames from the first quarter of the video and computes the pixel-wise median. This temporal median provides a stable reference frame in which brief occlusions are suppressed.

Grid regions are detected by applying Otsu thresholding to the grayscale reference frame, followed by heavy morphological closing (kernel size proportional to image dimensions) to merge all chambers within a panel into a single connected component. A subsequent morphological opening removes thin noise bridges between physically separate grids. Contours of the resulting blobs are filtered by area (at least 5% of the total frame area) and aspect ratio (between 0.3 and 3.0) to yield grid-level bounding boxes.

3.2 Grid Snapping and Regularization

Within each grid region, individual chambers are detected by a second round of Otsu thresholding and contour analysis at finer scale, with area-based filtering (0.5% to 25% of the grid region) and aspect-ratio constraints. The resulting candidate bounding boxes are validated for size consistency: detections whose area deviates by more than a factor of 2.5 from the median are rejected as outliers (debris, hands, shadows).

The remaining candidates are then snapped to a regular grid. The algorithm clusters candidate centers along the X and Y axes using the median box dimensions as a gap threshold. Row and column centers are computed from these clusters, and the cell size is set to the 75th percentile of detected dimensions (slightly larger than the median, to avoid clipping chamber edges that contour detection may have missed). A uniform grid of bounding boxes is reconstructed from these row/column centers and the standardized cell size, with configurable padding (default 15 pixels) to ensure complete coverage.

3.3 Video Stabilization via ECC

After cropping, each chamber video may optionally be stabilized to correct for camera vibration or slow arena drift. The stabilization module aligns every frame to the first frame using the Enhanced Correlation Coefficient (ECC) method of Evangelidis and Psarakis (2008), implemented in OpenCV as `cv2.findTransformECC()`. The ECC criterion is invariant to global photometric changes (brightness, contrast), making it suitable for experimental recordings with variable illumination.

Two optimizations are applied for computational efficiency. First, ECC alignment is computed on half-resolution images, reducing per-frame computation by approximately 4x; the resulting translation parameters are then scaled back to full resolution. Second, when the estimated motion magnitude falls below 0.5 pixels, the current warp matrix is reused for the next 5 frames without recomputing ECC. A Euclidean motion model (rotation plus translation) is used, with a

maximum of 20 iterations and a convergence threshold of 10^{-3} .

4. Fly Tracking

4.1 Background Subtraction (MOG2)

Foreground segmentation is performed using the Gaussian Mixture Model (MOG2) algorithm of Zivkovic (2004) and Zivkovic and van der Heijden (2006). Each pixel is modeled as a mixture of Gaussian distributions whose parameters are updated recursively as new frames are processed. The algorithm automatically adapts the number of mixture components per pixel, typically using between 3 and 5 components depending on the local complexity of the background. This adaptive behavior is well suited to courtship arenas where the background is largely static but may exhibit slight variations due to lighting changes or arena texture.

The MOG2 background subtractor is configured with a history window of 100 frames, a variance threshold of 20, and shadow detection disabled (as the white-background arenas used in courtship assays do not produce strong shadows). The resulting foreground mask is refined through morphological opening (3x3 elliptical kernel) to remove salt-and-pepper noise, followed by morphological closing (5x5 elliptical kernel) to fill small holes within fly body regions. A binary threshold at value 127 is applied to produce the final segmentation mask.

4.2 Ellipse Fitting and Body Modeling

External contours are extracted from the binary segmentation mask and filtered by area (minimum 0.05% of the chamber dimensions) to remove noise. The two largest valid contours (corresponding to the two flies) are retained. For each contour with at least 5 points, an ellipse is fit using the direct least-squares method of Fitzgibbon et al. (1999), as implemented in OpenCV's `cv2.fitEllipse()`. This yields the centroid (x, y), semi-major axis (a), semi-minor axis (b), and orientation angle for each fly.

The geometric wing angle is computed by subtracting the fitted body ellipse from the original fly contour, yielding a residual 'wings mask.' Connected components of this mask are filtered by size (between 5% and 50% of the body area) and proximity to the body centroid (within 2x the major axis length). The angle between the wing centroid and the body heading vector gives the wing extension angle.

4.3 Head-Tail Orientation Resolution

Ellipse fitting inherently carries a 180-degree ambiguity: the fitted orientation could point toward the head or the tail. This ambiguity is resolved through a combination of velocity-based and temporal-continuity cues.

For each frame, two candidate orientations (θ and $\theta + \pi$) are evaluated. When the fly is moving above a speed threshold (set adaptively at 20% of the median speed across the video), the candidate whose direction is closer to the instantaneous velocity vector is preferred. When velocity cues conflict with temporal continuity (the orientation from the previous frame), a hysteresis mechanism resolves the conflict: velocity is preferred only when the angular difference between the two candidates exceeds 0.3 radians; otherwise, the previous orientation is propagated. When the fly is stationary, temporal continuity alone determines the orientation.

This pixel-mass check is complemented at the detection level: for each frame, the foreground pixels within a region of interest around the fly centroid are projected along the ellipse's major axis. The half with greater cumulative intensity (after inverting the grayscale, so darker pixels carry more weight) is designated as the head direction. This two-stage

approach (pixel mass at detection time, velocity-based hysteresis at the trajectory level) provides stable orientation estimates even during stationary periods or brief occlusions.

4.4 Identity Assignment via the Hungarian Algorithm

Frame-to-frame identity assignment ensures that the same physical fly retains the same label throughout the video. This is formulated as a bipartite assignment problem: given the positions of n tracked flies in the previous frame and m detections in the current frame, find the assignment that minimizes total displacement.

The cost matrix is populated with Euclidean distances between each previous position and each current detection. The optimal assignment is found using the Hungarian algorithm (Kuhn, 1955; Munkres, 1957), via `scipy.optimize.linear_sum_assignment()`. A maximum distance threshold (set at 25% of the smaller chamber dimension) is enforced to prevent identity swaps when flies pass close to each other: matches exceeding this threshold are rejected, and the corresponding fly is marked as temporarily lost (NaN). When a fly is lost, its last known position is retained for matching in subsequent frames, enabling re-identification after brief occlusions.

4.5 Post-Processing: Gap Interpolation and Smoothing

After tracking, short NaN gaps (up to 5 frames) are interpolated. Positional features (x , y , semi-axis lengths, area) are linearly interpolated between the flanking valid frames. Orientation (θ) is interpolated using circular interpolation to avoid wraparound artifacts. Wing angle is set to zero during interpolated frames, as no reliable measurement is available.

A light Gaussian smoothing ($\sigma = 1.5$ frames) is applied to the x and y trajectories to reduce detection jitter without blurring fast movements. NaN regions are handled by temporary linear interpolation before smoothing, with NaN values restored afterward.

Overlap detection flags are computed by comparing the centroid-to-centroid distance to the sum of the semi-major axes of both flies. Frames in which this distance is smaller than the summed body lengths are flagged as overlap frames. Downstream classifiers use these flags to zero out behavioral scores during overlaps, preventing false positive detections when flies are in physical contact and individual features cannot be reliably measured.

5. Feature Extraction

5.1 Kinematic Features

Several kinematic features are computed from the tracked trajectories. All velocity computations apply Gaussian smoothing ($\sigma = 0.05 \times \text{fps frames}$) to the position traces prior to numerical differentiation, reducing amplification of detection noise by the gradient operator.

Forward and Lateral Velocity (v_{par} , v_{perp})

The velocity vector (v_x , v_y) is decomposed into components parallel and perpendicular to the fly's body axis. The forward (parallel) velocity v_{par} is the dot product of the velocity vector with the heading unit vector ($\cos(\theta)$, $\sin(\theta)$). The lateral (perpendicular) velocity v_{perp} is the dot product with the orthogonal vector ($-\sin(\theta)$, $\cos(\theta)$). These components distinguish between forward locomotion (relevant for following) and lateral movement (relevant for circling).

Angular Velocity

Angular velocity is computed as the frame-to-frame difference in orientation, wrapped to the interval $[-\pi, \pi]$ and multiplied by the frame rate to yield radians per second. Gaussian smoothing ($\sigma = 0.05 \times \text{fps}$) is applied after differentiation.

Total Speed

Total speed is the Euclidean magnitude of the smoothed velocity vector: $\sqrt{v_x^2 + v_y^2}$. This feature is used in copulation detection, where both flies are expected to be nearly stationary.

5.2 Social Geometry Features

Social features describe the spatial relationship between the focal fly and its partner. These are computed in the module `tracking/features.py`, function `compute_social_features()`.

Centroid-to-Centroid Distance ($c2c$)

The Euclidean distance between the centroids of the two flies. This is the primary proximity measure and enters the thresholds for following, copulation, and circling.

Facing Angle

The angle between the focal fly's heading direction and the bearing toward the partner's centroid, computed as bearing - θ and wrapped to $[-\pi, \pi]$. A facing angle near zero indicates the fly is oriented toward its partner; a value near π indicates it is facing away.

Nose-to-Ellipse Distance ($n2e$)

The distance from the focal fly's nose (tip of the body ellipse along the heading direction) to the surface of the partner's body ellipse. The partner ellipse's radius in the direction of the nose point is computed analytically using the formula

$r(\phi) = ab / \sqrt{(b \cos(\phi))^2 + (a \sin(\phi))^2}$, where ϕ is the angle relative to the partner's orientation. The $n2e$ distance is the raw distance from nose to partner centroid minus this directional radius. This feature is central to attempted copulation detection, where the male's nose must be in contact with the female's body.

Wing Angle

The geometric angle of the largest wing blob relative to the body axis, extracted during the tracking stage (Section 4.2). This feature drives wing extension detection.

5.3 JAABA-Compatible Window Features

When JAABA boosting classifiers are used, the system constructs a feature matrix compatible with JAABA's window feature specification. Each window feature is defined by a base per-frame time series (e.g., velocity, facing angle, wing angle), an optional transform (absolute value, flip, relative), a window statistic (mean, min, max, standard deviation, change, z-score of neighbors, difference from neighbor mean/min/max), a temporal radius, and a temporal offset.

The feature name tokens are parsed from the JAABA .jab file's classifier structure. A synonym map resolves naming variations between different JAABA versions and custom classifiers (e.g., 'speed' maps to 'vel', 'bearing' maps to 'bearing_to_other', 'facingangle' maps to 'facing_angle'). The relative transform bins values into percentile ranks, following JAABA's convention.

6. Behavior Classification

6.1 Heuristic (Rule-Based) Classifier

The heuristic classifier implements rule-based detectors for five courtship behaviors. Each rule computes a per-frame binary score from thresholded feature comparisons. The thresholds are set to default values derived from published behavioral criteria and can be adjusted through the Streamlit interface. All distance-based thresholds are specified in millimeters and converted to pixels using the user-supplied calibration factor (px_per_mm).

Behavior 1: Wing Extension

Wing extension is the hallmark of the courtship song. The male extends one wing to approximately 90 degrees from the body axis and vibrates it to produce a species-specific song composed of pulse and sine components (Greenspan and Ferveur, 2000).

Detection rule: A frame is scored positive when the geometric wing angle exceeds a threshold (default: 60 degrees). The default threshold is set below the full 90-degree extension to capture partial extensions that are still part of the courtship song. Bouts shorter than 1.0 second are rejected to suppress transient wing movements unrelated to courtship.

Behavior 2: Following

Following occurs when the male orients toward the female and moves toward her. It is one of the most common courtship behaviors and often occurs concurrently with wing extension (Hall, 1994).

Detection rule: A frame is scored positive when: (1) the forward (parallel) velocity v_{par} exceeds 2.0 mm/s; (2) the absolute facing angle toward the partner is less than 45 degrees; (3) the centroid-to-centroid distance is between 2.0 mm (minimum, to exclude copulation-range proximity) and 8.0 mm (maximum, to exclude independent locomotion far from the partner). The use of forward velocity (v_{par}) rather than total speed is deliberate: it ensures that only motion directed toward the partner is counted, excluding lateral or backward movements. Bouts shorter than 1.0 second are rejected.

Behavior 3: Circling

Circling is a lateral movement pattern in which the male moves around the female, maintaining proximity while oriented at an oblique angle. It typically occurs during close-range courtship.

Detection rule: A frame is scored positive when: (1) the absolute lateral (perpendicular) velocity exceeds 2.0 mm/s; (2) the centroid-to-centroid distance is less than 10.0 mm; (3) the absolute facing angle is between 45 degrees and 135 degrees (indicating the fly is oriented neither directly toward nor directly away from the partner, consistent with lateral circling). Bouts shorter than 1.0 second are rejected.

Behavior 4: Copulation

Copulation is characterized by sustained close contact between both flies with minimal movement. It is typically the longest-duration courtship behavior, lasting from several minutes up to approximately 20 minutes in *D. melanogaster*.

Detection rule: A frame is scored positive when: (1) the centroid-to-centroid distance is less than 2.0 mm; (2) the fly's total velocity is less than 0.3 mm/s; (3) the rolling standard deviation of the centroid-to-centroid distance over a 2-second

window is less than 0.5 mm, ensuring the close contact is spatially stable rather than a momentary overlap. Bouts shorter than 8.0 seconds are rejected, consistent with the known minimum duration of copulation.

Behavior 5: Attempted Copulation

Attempted copulation occurs when the male's head makes contact with the female's body while he is still in motion -- typically involving mounting attempts that do not result in sustained copulation.

Detection rule: A frame is scored positive when: (1) the nose-to-ellipse distance (n2e) is less than 0.5 mm (indicating physical contact between the male's head and the female's body); (2) the total velocity exceeds 1.0 mm/s (distinguishing active mounting attempts from the static posture of actual copulation). Bouts shorter than 0.5 seconds are rejected.

The complete set of default parameters is summarized in the following table:

Behavior	Parameter	Default	Unit
Wing Extension	Angle threshold	60	degrees
Wing Extension	Min duration	1.0	seconds
Following	Forward velocity	2.0	mm/s
Following	Facing angle	45	degrees
Following	Distance range	2.0-8.0	mm
Following	Min duration	1.0	seconds
Circling	Lateral velocity	2.0	mm/s
Circling	Max distance	10.0	mm
Circling	Facing angle range	45-135	degrees
Circling	Min duration	1.0	seconds
Copulation	Max distance	2.0	mm
Copulation	Max velocity	0.3	mm/s
Copulation	Stability (c2c std)	0.5	mm
Copulation	Min duration	8.0	seconds
Attempted Cop.	Nose distance	0.5	mm
Attempted Cop.	Min velocity	1.0	mm/s
Attempted Cop.	Min duration	0.5	seconds
General	Merge gap	0.5	seconds

6.2 JAABA Boosting Classifier

As an alternative to the heuristic classifier, the system supports JAABA-trained GentleBoost classifiers (Kabra et al., 2013; Friedman et al., 2000). JAABA models are stored in .jab files, which are MATLAB v7 or v7.3 (HDF5) archives containing the trained weak learners (decision stumps) and the feature specification.

Each weak learner is parameterized by four values: (1) dim, the index of the feature it evaluates (1-based, following MATLAB convention); (2) thr, the threshold; (3) direction, indicating which side of the threshold corresponds to the positive class; and (4) alpha, the weight of this learner in the ensemble. The strong classifier score for each frame is the sum of all weak learner contributions:

$$\text{score}(t) = \sum_k [(2 * I(x[\text{dim}_k](t) < \text{thr}_k) - 1) * \text{dir}_k * \alpha_k]$$

where $I(.)$ is the indicator function. Frames with positive total score are classified as exhibiting the behavior. The .jab

parser supports both MATLAB v7 format (via `scipy.io.loadmat`) and v7.3 HDF5 format (via `h5py`), and handles nested structures, string/byte encoding variations, and sidecar token files for feature name specification.

6.3 Temporal Smoothing and Bout Detection

Raw binary classification scores are temporally smoothed before bout extraction. A Gaussian filter ($\sigma = 0.05 \times \text{fps}$ frames, typically 1-2 frames at 30 fps) is applied to the binary score array, and the result is re-thresholded at 0.5. This eliminates isolated single-frame false positives while preserving the boundaries of genuine bouts.

Bouts are extracted by identifying contiguous runs of positive scores. Each run must meet a minimum duration threshold (specific to each behavior, as described above). After initial extraction, a merge step combines bouts separated by gaps shorter than a configurable merge threshold (default: 0.5 seconds). This prevents a single courtship episode from being fragmented into many short bouts by brief score fluctuations. After merging, the minimum duration filter is applied again to reject any merged bouts that still fall below the threshold.

6.4 Mutual Exclusion and Priority Resolution

Some behaviors are physically incompatible and should not co-occur in the same frame. The system enforces mutual exclusion through a priority hierarchy:

Copulation > Attempted Copulation > Following > Circling

Wing Extension is exempt from mutual exclusion, as it can co-occur with following or circling (the male often sings while following). For the remaining four behaviors, frames are claimed in priority order: once a frame is assigned to a higher-priority behavior, it cannot be claimed by a lower-priority one. Bouts that lose all their unclaimed frames are removed.

7. Output and Summary Statistics

7.1 Courtship Index

The Courtship Index (CI) is computed following the definition of Siegel and Hall (1979): the percentage of the total observation time during which the fly engages in any courtship behavior. The implementation creates a boolean array spanning all frames, marks frames belonging to any detected bout (across all five behaviors and all flies), and computes $CI = (\text{marked frames} / \text{total frames}) * 100$. The CI is reported separately for each fly.

7.2 Sex Inference

When sex labels are not provided a priori, the system infers sex from behavioral asymmetry. The fly that accounts for more than 70% of the total wing extension duration is classified as male, and the other as female. This ratio-based criterion reflects the strong sexual dimorphism in courtship song production: males perform the vast majority of wing extensions in a typical courtship assay. If neither fly exceeds the 70% threshold (as may occur when both flies are male or when courtship is minimal), no sex assignment is made.

7.3 CSV Export Format

Two CSV files are produced per chamber:

Detailed results (chamber_N_results.csv): One row per behavior bout, with columns for Behavior, Fly, Start_Time (s), End_Time (s), Duration (s), Start_Frame, and End_Frame. Rows are sorted chronologically.

Summary statistics (chamber_N_summary.csv): Per-fly metrics in a long-format table with columns Fly, Metric, and Value. Metrics include per-behavior bout counts and durations, Courtship Index (%), Latency to First Courtship (s), Latency to Copulation (s), and Video Duration (s).

8. Installation and Usage

8.1 System Requirements

- Python 3.9 or later
- A video file of a multi-chamber courtship arena (MP4, AVI, MOV, or MKV)
- Sufficient RAM for parallel chamber processing (approximately 0.5 GB per chamber)
- Optional: FFmpeg installed and available on PATH for faster video cropping
- Optional: JAABA .jab classifier files for boosting-based classification

8.2 Installation Steps

1. Clone or download the project repository:

```
git clone <repository-url>
cd Courtship_Analysis_Python
```

2. Create and activate a virtual environment (recommended):

```
python -m venv venv
# On Windows:
venv\Scripts\activate
# On macOS/Linux:
source venv/bin/activate
```

3. Install the required dependencies:

```
pip install -r requirements.txt
```

The requirements.txt file specifies the following packages:

```
streamlit>=1.30.0
opencv-python-headless
numpy>=1.24.0
scipy>=1.10.0
pandas>=2.0.0
ffmpeg-python>=0.2.0
h5py>=3.10.0
matplotlib>=3.7.0
Pillow>=9.0.0
psutil>=5.9.0
```

8.3 Running the Application

Launch the Streamlit web application:

```
streamlit run app.py
```

This will start a local web server (typically at <http://localhost:8501>) and open the application in a browser. The interface

consists of two tabs:

- Analysis tab: Upload or specify a video file, detect chambers, adjust parameters, select chambers, and run the full analysis pipeline.
- Verification tab: Browse completed analysis results, view GIF previews of detected bouts, inspect behavior timelines, and review summary statistics.

8.4 Analysis Workflow

Step 1: Provide the input video via file upload or by entering a local file path.

Step 2: Review the auto-detected chambers overlaid on the first frame. Adjust using the Manual Grid Builder or the raw ROI editor if needed.

Step 3: Set the calibration factor (pixels per mm) based on a known dimension in the arena. This is required for the heuristic classifier's distance-based thresholds.

Step 4: Choose the analysis method (Heuristic or JAABA Classifiers). If using heuristic mode, threshold parameters can be adjusted in the sidebar expander.

Step 5: Select which chambers to process and whether to enable video stabilization.

Step 6: Click 'Start Analysis.' Progress is displayed in real time, and results appear incrementally as chambers complete. Processing is parallelized across available CPU cores (with RAM-aware worker count adjustment).

Step 7: Download individual CSV files or a ZIP archive of all results.

8.5 Resumable Processing

The system saves progress after each chamber completes. If the analysis is interrupted (browser closure, system crash), re-running on the same video will detect previously completed chambers and skip them, processing only the remaining chambers.

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